

Civil Engineering – Case Study

Prof. Ortega is working the civil engineering department of a large university campus, studying real-time traffic signal performance measures in which he observes the movement of traffic, specifically the number of vehicles passing through an intersection and the amount of time they spend at an intersection over a 24 hour period. He uses this information to generate “profiles” of traffic movement at intersections. These profiles also take into consideration contextual elements including weather conditions, local zoning or the surrounding area (heavy commercial, light residential, etc.) and the condition of the roads. His objective is to improve traffic conditions through urban areas by studying and accounting for factors that either improve or impede traffic flow. He works closely with the state department of transportation (his major source of funding), and his data is used to make decisions on public works spending and as a means of informing public policy.

Ortega has several graduate students working in his lab, all of whom are using different naming conventions for the data they use in their spreadsheets. Researchers on the project have complained that they can’t always make sense of the data that is sent to them, since no metadata or explanatory information is included in the files, and the field names are unclear.

Ortega’s graduate students are all over the map in terms of their skills and abilities in handling data. A few of them do a nice job of tracking the work that they do and explaining how and why they do things, others do not. Most of his students make some annotations in the data to explain perceived anomalies, but their annotations vary in their descriptive quality. He does try to guide his students during their one-on-one meetings, but admits that he does not always have the time to review the students’ work with the data with as much depth as he would like. Some students are using a file naming convention that appends dates to file names to indicate recency, others are trying a versioning system, by appending a version number to the end of the file name. One particularly ambitious student has started an Excel document that lists all of these file names, with a statement about what is contained within the file in the next column over. Although, unfortunately sometimes all that is listed is ???.

He delivers a copy of his data to the state department of transportation as several zipped packages over an FTP server; they do not ask him to follow a particular standard or format.

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The state will occasionally ask some questions about the data, which he will answer or refer to his students. Prof. Ortega believes that his data could be very interesting and potentially very useful to the general public, and would share the data if the state would allow it (he hasn't asked what their policies are). He also needs information about the appropriate venue and process for sharing his data.

To start the data collection, Prof. Ortega places sensors at each lane of traffic at an intersection. These sensors produce a raw data stream of data measuring traffic flow in a format that is proprietary to the sensor. The raw data is exported into a .csv files which are then cleaned up and added into a MySQL database. Prof. Ortega's lab supports multiple MySQL databases and each one holds between 3-4 months' worth of data. Data are extracted from the MySQL database into MS Excel for creating graphs and charts. Other components of the data set include videos taken of traffic flows at the intersection, a database of road conditions maintained by the state transportation department and data about weather conditions that may be needed to explain deviations in data points from what was expected.

Right now the data is stored on the lab's centralized server, and each of the students has access to the lab, which is kept locked after hours and on weekends. No one is too concerned about privacy, since personal information isn't captured, just raw counts of vehicles. Backups are handled with whatever procedures the department's IT personnel are using (Prof. Ortega isn't sure of the specifics). As previously stated, Prof. Ortega would like help figuring out if he can share his data, including where to share it, so that others might benefit from his research. He'd also like to archive his data somewhere long-term, since the time study may also prove useful to future research.